Eslam El-Saied Hosainy

AI ENGINEER

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LinkedIn: Eslam El-Saied | LinkedIn | Portfolio: Eslam Elsaied | ML Engineer & Data Analyst

Summary

AI Engineer specializing in **Machine Learning** and **Data Analysis** using **Python** and modern ML libraries. Experienced in handling **large-scale datasets (50,000+ records)**, performing **Data Preprocessing**, **Feature Engineering**, and **Exploratory Data Analysis** to build reliable predictive models. Achieved **high-performance results**, including a **0.992 R^2 score on Kaggle benchmarks**. Skilled in developing **data-driven AI solutions** that transform complex data into **actionable insights and practical applications**.

Education

Misr Higher Institute for Commerce & Computers in El-Mansoura | Faculty of Computer Science
Bachelor's in Computer Science | Expected Graduation: 2026

Experience

Digital Egypt Pioneers Initiative (DEPI) November 2025 – Present
Machine Learning Engineer (Internship)

- **Developed and deployed 5+ Machine Learning models** for classification and regression tasks using Python and Scikit-learn, achieving an average accuracy of 92%.
- **Engineered robust feature pipelines** and performed extensive data preprocessing on datasets with 30,000+ records to enhance model generalization.

National Telecommunication Institute (NTI) November 2025 – February 2026

Machine Learning for Data Analysis (Internship)

- **Performed comprehensive Exploratory Data Analysis (EDA)** on datasets with 20,000+ records, uncovering 10+ key business trends using **Pandas, Matplotlib, and Seaborn**.
- **Developed automated data cleaning and feature engineering scripts** that handled missing values and outliers, improving data quality and reducing model error rates by **~10%**.
- **Applied unsupervised learning (K-Means Clustering)** to segment customer behavior, identifying 5 distinct user groups and providing actionable insights for targeted marketing strategies.

Technical Skills

Programming Languages: Python **(Expert)**, C++

AI and Machine Learning: Machine Learning (Supervised/Unsupervised), Feature Engineering, Model Evaluation

Data Analysis and Visualization: Exploratory Data Analysis (EDA), Statistical Analysis

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Tools & Platforms: VS Code, Jupyter Notebook, Google Colab

Soft Skills: Problem Solving, Analytical Thinking, Teamwork, Business Communication

Projects

Agrofood CO2 Emissions Prediction & Analysis

(Tech: Python, Streamlit, Random Forest, Scikit-learn, Pandas, Seaborn, Matplotlib)

- **Collaborated with a technical team** to achieve a **top-tier R^2 score of 0.992** using Random Forest Regressor, surpassing existing public benchmarks on Kaggle for this dataset.
- **Led the data preprocessing and feature engineering** phases, handling a complex dataset of **7,000+ records and 31 features** to ensure maximum model reliability and performance.
- Developed an **interactive Web Dashboard using Streamlit**, enabling users to visualize global emission trends and obtain real-time predictions through user-defined variables.
- Utilized **Matplotlib and Seaborn** to create advanced visualizations, identifying and presenting key factors contributing to global food system emissions.

Certifications

- **Introduction to AI & Generative AI** – Microsoft | **October 2025**
- **Business English** – Mansoura University (DTC) | **October 2025**
- **Data Science and Machine Learning** – Mansoura University (DTC) | **September 2025**
- **Python Programming Language** – Microsoft | **August 2025**
- **AI and Machine Learning** – Sprints x Microsoft Summer Camp | **2025**
- **Full Stack Web Developer** – Mansoura University (DTC) | **September 2024**
- **Soft Skills** – Mansoura University (DTC) | **September 2024**